# Linear Regression

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# Outline


The Linear Model
- Simple linear regression
- History
- Multiple linear regression

Solution methods
- Gradient Descent
- Least-Squares

Regression libraries in Python
- sklearn
- statsmodels

Problems
- Interpretation Problem parameters
- Exercises

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# Simple linear regression

## Input and output

- Data pairs $(x^t, y^t)$, $t = 1, \ldots, T$.
- Input $x^t \in \mathbb{R}$
- Output $y^t \in \mathbb{R}$.

## Modelling the conditional expectation $\mathbb{E}[y^t|x^t]$

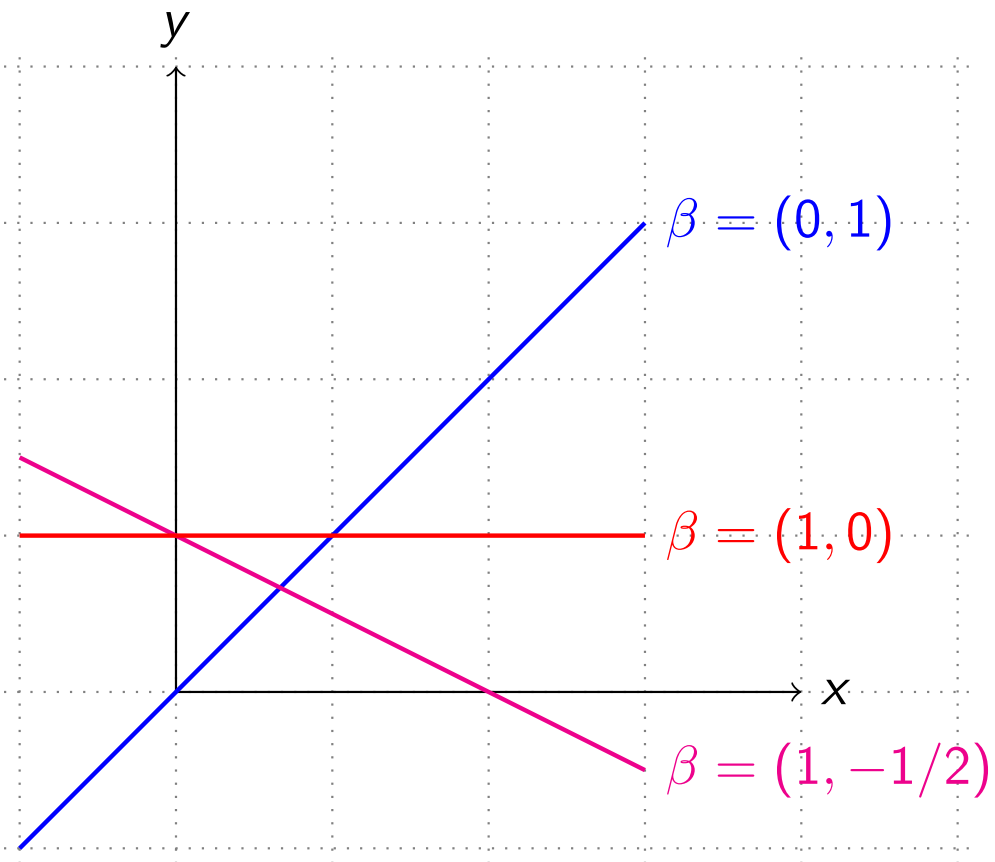
- Parameters $\theta_0, \theta_1 \in \mathbb{R}$
- Function $\pi_\theta : \mathbb{R} \to \mathbb{R}$, defined as

$$\pi_\theta(x^t) = \theta_0 + \theta_1 x_t$$

## Probabilistic predictions: Modelling the conditional probability $\mathbb{P}[y^t|x^t]$

- $y^t = \mathbb{E}[y^t|x^t] + \epsilon_t$, with $\epsilon_t \in \mathbb{R}$ being zero-mean noise.
- Simplest model: variance $\sigma = \mathbb{V}(\epsilon) = \mathbb{E}[\epsilon_t^2]$

# Linear models




$$\pi_\theta(x) = \theta_0 + \theta_1 x = [\theta_0, \theta_1] \begin{bmatrix} 1 \\ x \end{bmatrix}$$

# Two views of the problem

## Learning as optimisation

- Each value $\pi_\theta(x)$ is a prediction about the value of $y$

## Learning as inference

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## Learning as optimisation

- Each value $\pi_\theta(x)$ is a prediction about the value of $y$
- We suffer a loss $\ell(y, \pi_\theta(x))$ for every example $(x, y)$ that we see.

## Learning as inference

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- Each value $\pi_\theta(x)$ is a prediction about the value of $y$
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- We can minimise the average loss over the data

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## Learning as inference

- The parameters $\theta$ define a probabilistic model $P_\theta(y|x)$ for every value of $y$.

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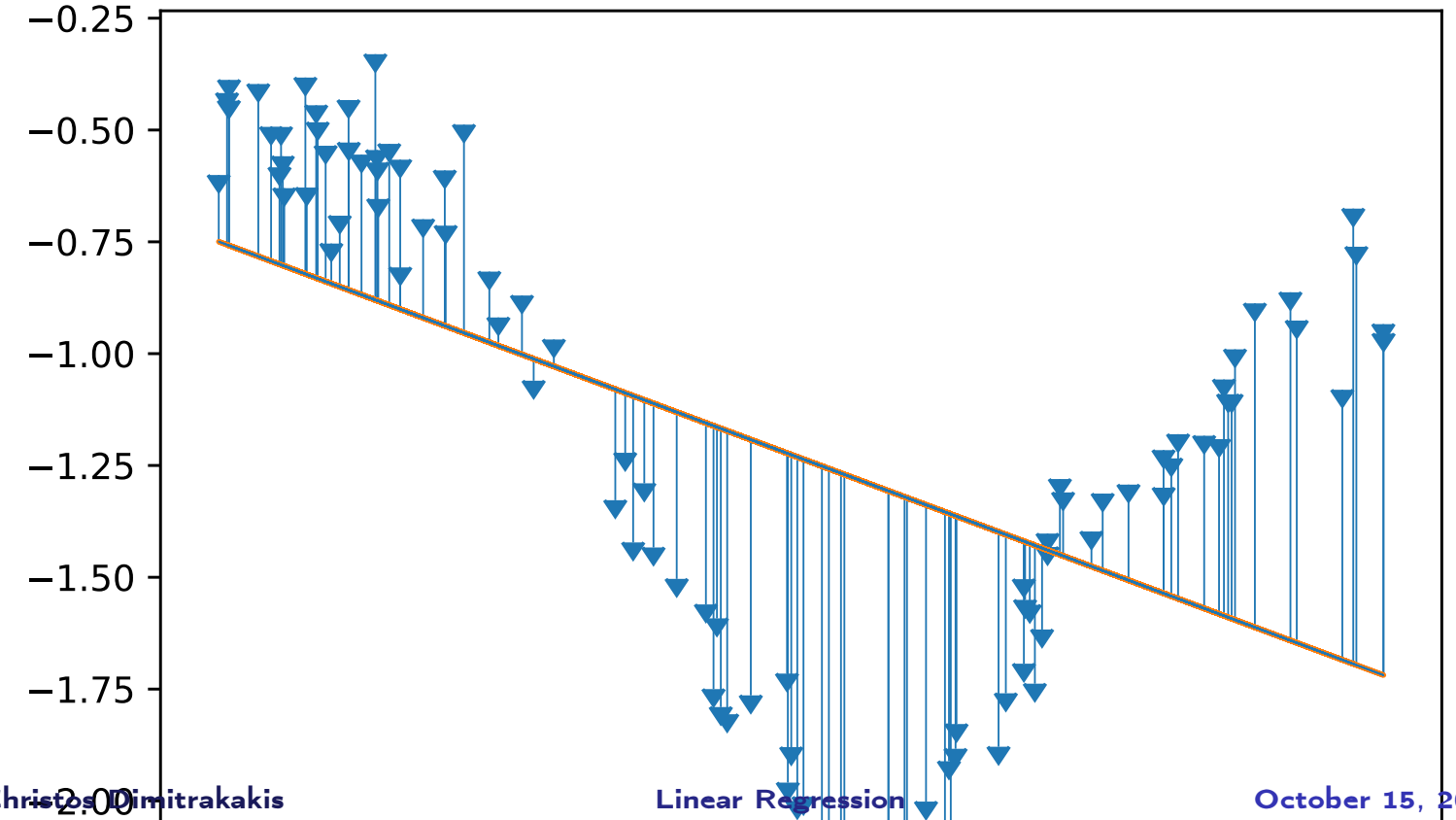
## Learning as inference

- The parameters $\theta$ define a probabilistic model $P_\theta(y|x)$ for every value of $y$.
- We want to find the parameters giving the highest probability on the observed data.
- Ideally, we want to find the true conditional distribution $P(y|x)$.

# Learning as Optimisation

Find the parameters $\theta$ minimising squared error

$$\min_{\theta} \frac{1}{T} \sum_{t=1}^{T} \Big[ \underbrace{y^t - \pi_\theta(x^t)}_{\text{residual}} \Big]^2$$



# Origins

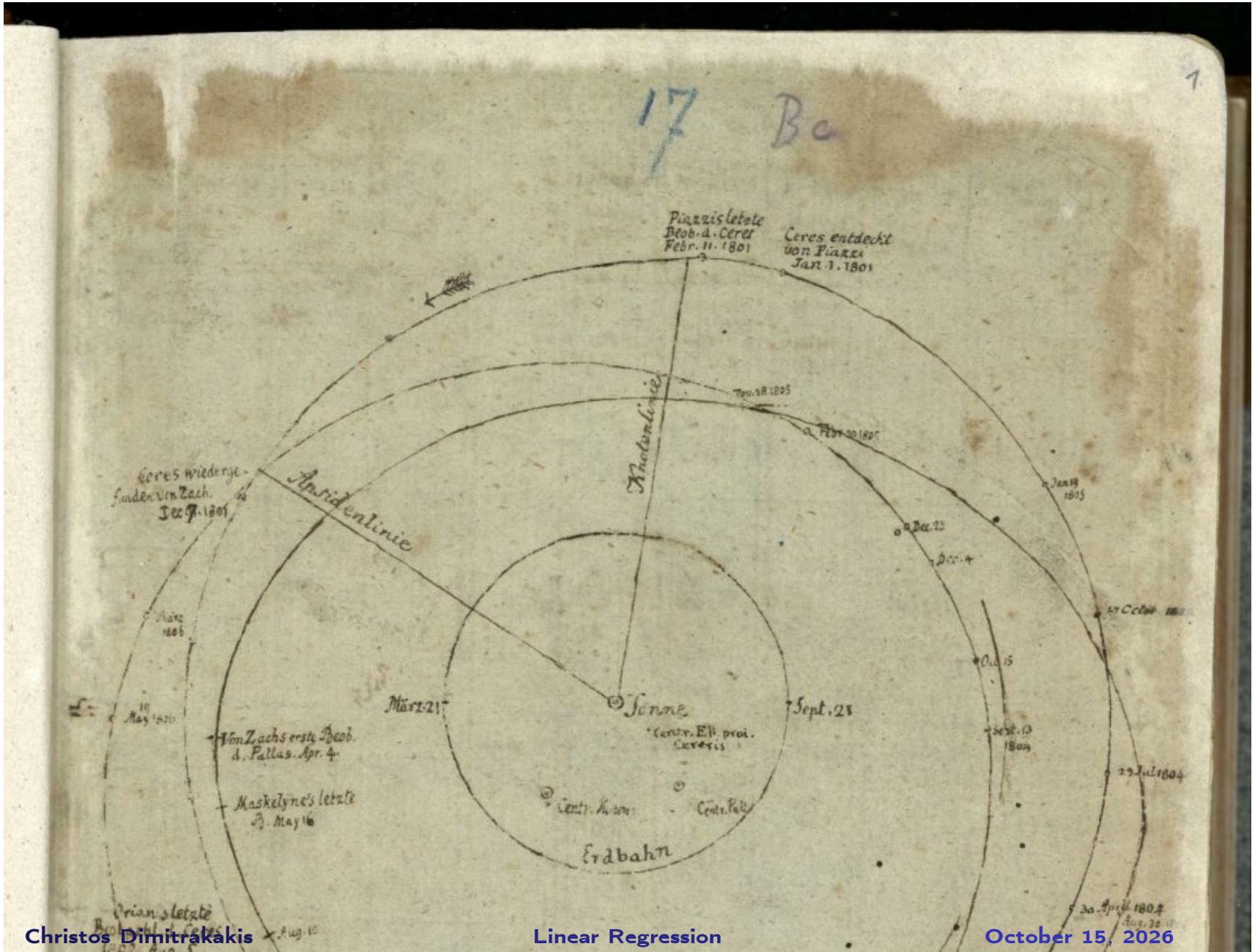


Figure: Gauss: originator

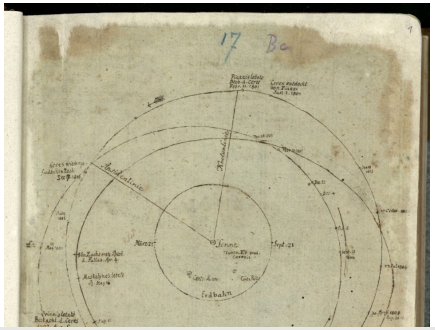


Figure: Legendre: first publication

# The orbit of Ceres



The orbit of Ceres



On Gauss and Legendre. Gauss applied the method of least squares to calculate the orbit of Ceres, an asteroid detected in the solar system. At the time, astronomers were keen to establish what the orbit was, but they did not even know where to point their telescopes. He was the only one who predicted the orbit sufficiently well to tell them where to look. He apparently had already developed the method when he was aged 18, but he only published it much later. He also, of course, famously defined the normal (Gaussian) distribution, which is closely linked to the least-squares method.

The full story of Gauss, Laplace, Legendre and Ceres.

# Maximum likelihood inference

## Gaussian noise model.

$$y^t = \pi(x^t) + \epsilon_t, \qquad \epsilon_t \sim \text{Normal}(0, \sigma)$$

With conditional density:

$$p_\theta(y^t|x^t) \propto \exp(-[y^t - \pi_\theta(x^t)]^2/2\sigma^2)$$

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## Maximum likelihood inference: Find parameters maximising $P_\beta(D)$

$$\begin{aligned}
\arg\max_\theta P_\beta(D) &= \arg\max_\theta p_\theta(y^1, \ldots, y^t|x^1, \ldots, x^t) = \arg\max_\theta \ln \prod_t p_\theta(y^t|x^t) \\
&= \arg\max_\theta \sum \ln p_\theta(y^t|x^t) \\
&= \arg\max_\theta \sum_t \ln \left\{\exp\left(-[y^t - \pi_\theta(x^t)]^2/2\sigma^2\right)\right\} \\
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Gaussian noise model. In regression, we can never exactly predict $y^t$. In terms of loss minimisation, we are simply trying to minimise some measure of error in our predictions. However, an alternative approach is to assume that $y^t$ is inherently unpredictable, but that it follows a specific distribution, with a mean centered around the prediction $\pi_{\boldsymbol{\theta}}(x^t)$. Mathematically, we express this by saying that $y^t$ is the result of adding some noise $\epsilon_t$ to the output of the mean function $\pi(x^t)$,

$$y^t = \pi_\theta(x^t) + \epsilon_t.$$

Since we have to assume something about the noise, a common assumption is that it is normally distributed. We write this as

$$\epsilon_t \sim \text{Normal}(0, \sigma),$$

where $\text{Normal}(0, \sigma)$ means the normal distribution with mean 0 and variance $\sigma^2$. We can also write this in terms of a conditional density function.

$$p_\theta(y^t|x^t)i \propto \exp(-[y^t - \pi_\theta(x^t)]^2/2\sigma^2),$$

where the proportionality symbol hides the constant $1/\sqrt{2\pi\sigma^2}$.

While the Gaussian noise model is not the only possibility, it has some interesting properties.

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Maximum likelihood. Now that we have a probabilistic model, we can assign probabilities to the data we have. This is the central idea of the maximum likelihood principle:

- For data $D$, find parameters $\beta$ maximising $P_\beta(D)$.
- Let us apply this principle to our case, and show that maximising the likelihood is the same as minimising the squared errors, when the noise is Gaussian.

Proof details. Here, first of all, we only care about predicting $y$ from $x$. For that reason, we define

$$P_\beta(D) p_\theta(y^1, \ldots, y^t|x^1, \ldots, x^t)$$

We also assume that $y^t$ only depends on $x^t$ if we know the mean and variance, so that $p_\theta(y^1, \ldots, y^t|x^1, \ldots, x^t) = \prod_t p_\theta(y^t|x^t)$.
The maximum point of some function $f(\theta)$ is the same of some monotonically increasing transformation $g$ of it, i.e. if $f(\theta^*) \geq f(\theta)$, then $g[f(\theta^*)] \geq g[f(\theta)]$. Hence, we can maximise $\ln \prod_t p_\theta(y^t|x^t)$ instead. And since $\ln(a+b) = \ln(a)\ln(b)$.

Finally, note that if $\exp(a) = b$ then $a = \ln(b)$.

# Coding break

src/Regression/RegressionDemonstration.ipynb

- Show implementation
- Fit and residuals
- Multiple draws from the distribution
- Fit on non-linear data?

# Multiple linear regression

## Input and output

- Data pairs $(\boldsymbol{x}^t, y^t)$, $t = 1, \ldots, T$.
- Input $\boldsymbol{x}^t \in \mathbb{R}^n$
- Output $\boldsymbol{y}^t \in \mathbb{R}^m$.

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## Point predictions: Modelling the conditional expectation $\mathbb{E}[y^t|x^t]$

- Parameters $\boldsymbol{\Theta} \in \mathbb{R}^{n \times m}$
- Function $\pi_{\boldsymbol{\Theta}} : \mathbb{R}^n \to \mathbb{R}^m$, defined as

$$\pi_{\boldsymbol{\Theta}}(x^t) = \boldsymbol{\Theta}^\top \boldsymbol{x}^t = \sum_{i=1}^{n} \boldsymbol{\theta}_i^\top x_i^t$$

## Probabilistic predictions: Modelling the conditional probability $\mathbb{P}[y^t|x^t]$

- $y^t = \mathbb{E}[y^t|x^t] + \epsilon_t$, with $\epsilon_t \in \mathbb{R}^m$ being zero-mean noise.
- Noise covariance matrix $\Sigma = \mathbb{V}(\epsilon) = \mathbb{E}[\epsilon_t \mid \epsilon_t \top]$

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Multiple linear regression Here, both the input and output are multi-dimensional. For that reason, we need an $n \times m$ matrix $\boldsymbol{\Theta}$ to predict the mean.
In fact, since the output has $m$ dimensions, and since the input has $n$, then

$$\boldsymbol{y} = \boldsymbol{\Theta}^\top \times \boldsymbol{x}$$

corresponds to a multiplication between matrices of the form

$$[m, 1] = [m, n] \times [n, 1].$$

The other thing that changes is that the noise may be correlated between different output dimensions. For that reason, we are using a multivariate Gaussian distribution, with mean $\boldsymbol{\Theta}^\top \times \boldsymbol{x}$ and covariance $\Sigma^\top \Sigma$.

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General comments. Now that we have shown that both formalisations of the problem lead to the same objective, under some conditions, let us see how we can actually find solutions.

The first idea is to use gradient descent. This allows us to find the minimum of any convex function, and a local mimimum of a differentiable function.

The second idea is to take a look at an analytical solution to the problem of finding the parameters $\boldsymbol{\theta}$ from the $\boldsymbol{x}, y$ pairs, by taking advantage of matrix algebra. Interestingly, the analyical solution turns out to be quite simple to express, and (when it exists) is the same as the least-squares solution.

# Gradient descent algorithm

## Minimising a function

$$\min_{\theta} f(\theta) \leq f(\theta') \forall \theta', \qquad \theta^* = \arg\min_{\theta} f(\theta) \Rightarrow f(\theta^*) = \min_{\theta} f(\theta)$$

# Gradient descent algorithm

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## Gradient descent for minimisation

- Input $\theta_0$
- For $n = 0, \ldots, N$:
- $\theta_{n+1} = \theta_n - \alpha_n \nabla_\theta f(\theta_n)$

# Gradient descent algorithm

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## Step-size $\alpha_n$

- $\alpha_n$ fixed: for online learning
- $\alpha_n = c/[c+n]$ for asymptotic convergence
- $\alpha_n = \arg\min_\alpha f(\theta_n + \alpha\nabla_\theta)$: Line search.

Gradient descent algorithm

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Gradient descent notes. Gradient descent can be used to minimise a convex function, or find a local minimum of a differentiable function. The idea is to simply move in the opposite direction of the gradient, i.e. the direction of steepest descent.
How much do we need to move? The step size $\alpha$ at every iteration can be fixed. This usually works reasonably well in practice. However, in theory, $\alpha$ must decrease over time.

Another alternative is line search: this minimises the function along the chosen direction.

The usual algorithm for doing this is binary search.

# Gradient descent for squared error

## The cost function

$L(\theta, D) = \sum_{t=1}^{T}(y^t - \pi_{\boldsymbol{\theta}}(x^t))^2 = \sum_{t=1}^{T} \epsilon_t^2$, with $\epsilon_t \triangleq y^t - \pi_{\boldsymbol{\theta}}(x^t)$.

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## Cost gradient

Using the chain rule of differentiation, $\nabla_{\boldsymbol{\theta}} f(\epsilon) = \nabla_{\epsilon} f(\epsilon) \nabla_{\boldsymbol{\theta}} \epsilon$.

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$$= \sum_{t=1}^{T} 2\epsilon_t \nabla_{\boldsymbol{\theta}} [y^t - \pi_{\boldsymbol{\theta}}(x^t)] = -\sum_{t=1}^{T} 2[y^t - \pi_{\boldsymbol{\theta}}(x^t)] \nabla_{\beta} \pi_{\boldsymbol{\theta}}(x^t)$$

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$$= \sum_{t=1}^{T} 2\epsilon_t \nabla_{\boldsymbol{\theta}}[y^t - \pi_{\boldsymbol{\theta}}(x^t)] = -\sum_{t=1}^{T} 2[y^t - \pi_{\boldsymbol{\theta}}(x^t)]\nabla_{\beta}\pi_{\boldsymbol{\theta}}(x^t)$$

## Parameter gradient for linear regression

Remember $\nabla_{\boldsymbol{\theta}} = (\frac{\partial}{\partial\theta_1}, \cdots, \frac{\partial}{\partial\theta_n})$ and consider the individual factors $\frac{\partial}{\partial\theta_j}\pi_{\boldsymbol{\theta}}(x^t)$:

$$\frac{\partial}{\partial\theta_j}\pi_{\boldsymbol{\theta}}(x^t) = \frac{\partial}{\partial\theta_j}\sum_{i=1}^{n}\theta_i x_i^t = \sum_{i=1}^{n}\frac{\partial}{\partial\theta_j}\theta_i x_i^t = x_j^t.$$

Gradient descent for squared error
The cost function
$L(\theta, D) = \sum_{t=1}^{T}(y^t - \pi_{\boldsymbol{\theta}}(x^t))^2 = \sum_{t=1}^{T} \epsilon_t^2$, with $\epsilon_t \triangleq y^t - \pi_{\boldsymbol{\theta}}(x^t)$.
Cost gradient
Using the chain rule of differentiation, $\nabla_{\boldsymbol{\theta}} f(\epsilon) = \nabla_\epsilon f(\epsilon) \nabla_{\boldsymbol{\theta}} \epsilon$.

$$\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}, D) = \nabla_{\boldsymbol{\theta}} \sum_{t=1}^{T} \epsilon_t^2 = \sum_{t=1}^{T} \nabla_{\boldsymbol{\theta}} \epsilon_t^2 = \sum_{t=1}^{T} \nabla_{\epsilon_t} \epsilon_t^2 \nabla_{\boldsymbol{\theta}} \epsilon$$
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Parameter gradient for linear regression
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Squared error notes. In our setting, we have loss function that is the sum of squared residuals, i.e. the squared error. To minimise the gradient, we use the chain rule of differentiation.This results in two parts. The error $\epsilon_t$ and the gradient of the output with respect to the parameters $\nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}$. For a linear function, this has a really simply form: $\partial / \partial \theta_j \pi_{\boldsymbol{\theta}}(\boldsymbol{x}) = x_j$.
Consequently, the batch gradient descent parameter update for linear regression will have the form

$$\boldsymbol{\theta}^{n+1} = \boldsymbol{\theta}^n + \alpha_n \sum_{t=1}^{T} [y^t - \pi_{\boldsymbol{\theta}}(x^t)] \nabla_\beta \pi_{\boldsymbol{\theta}}(x^t) \qquad \text{(vector form)}$$

$$\theta_i^{n+1} = \theta_i^n + \alpha_n \sum_{t=1}^{T} [y^t - \pi_{\boldsymbol{\theta}}(x^t)] x_i^t, \forall i \in [n]. \qquad \text{(component form)}$$

Here we dropped the constant 2 term, and folded it into the step size.

# Stochastic gradient descent algorithm

## When $f$ is an expectation

$$f(\theta) = \int_X dP(x) g(x, \theta).$$

## Replacing the expectation with a sample:

$$\begin{aligned} \nabla f(\theta) &= \int_X dP(x) \nabla g(x, \theta) \\ &\approx \frac{1}{K} \sum_{k=1}^{K} \nabla g(x^{(k)}, \theta), \qquad x^{(k)} \sim P. \end{aligned}$$

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Considering the distribution. Normally, we care about minimising the expected loss over the true data distribution $P(x, y) = P(y|x)P(x)$. However, we can get an unbiased estimate of the gradient by considering samples $(x, y) \sim P$ instead. This leads to a stochastic gradient descent algorithm.

# Analytical Least-Squares Solution

We need to solve the following equations for $A$:

$$\begin{aligned} y^1 &= (\boldsymbol{x}^1)^\top \boldsymbol{\theta} \\ \cdots &\quad \cdots \\ y^t &= (\boldsymbol{x}^t)^\top \boldsymbol{\theta} \\ \cdots &\quad \cdots \\ y^T &= (\boldsymbol{x}^t)^\top \boldsymbol{\theta} \end{aligned}$$

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We can rewrite it in matrix form:

$$\begin{pmatrix} y_1 \\ \vdots \\ y_t \\ \vdots \\ y_T \end{pmatrix} = \begin{pmatrix} \boldsymbol{x}_1^\top \\ \vdots \\ \boldsymbol{x}_t^\top \\ \vdots \\ \boldsymbol{x}_T^\top \end{pmatrix} \boldsymbol{\theta}$$

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Resulting in

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How can we get $\boldsymbol{\theta}$?

# Finding the $\theta$

We now have a linear equation,

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\theta}.$$

We want to solve for $\theta$. If $\boldsymbol{X}$ had an inverse $\boldsymbol{X}^{-1}$, we could obtain

$$\boldsymbol{X}^{-1}\boldsymbol{y} = \boldsymbol{X}^{-1}\boldsymbol{X}\boldsymbol{\theta} = \boldsymbol{I}\boldsymbol{\theta} = \boldsymbol{\theta}.$$

But $\boldsymbol{X}^{-1}$ does not exist.

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## Least-squares solution

The left-pseudo inverse $\tilde{\boldsymbol{X}}^{-1} \triangleq (\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top$ can be used to obtain

$$\boldsymbol{\theta} = \tilde{\boldsymbol{X}}^{-1}\boldsymbol{y},$$

This follows as:

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\boldsymbol{y} &= \boldsymbol{X}\boldsymbol{\theta} \\
\tilde{\boldsymbol{X}}^{-1}\boldsymbol{y} &= \tilde{\boldsymbol{X}}^{-1}\boldsymbol{X}\boldsymbol{\theta} \\
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Note on least squares. $\boldsymbol{X}^{-1}$ does not exist, because only square matrices can have inverses, and $\boldsymbol{X}$ is $T \times n$ in size.

This pseudo-inverse solution requires us to invert the matrix $(\boldsymbol{X}^\top \boldsymbol{X})$. Since $\boldsymbol{X}$ is $T \times n$, we are multiplying a $n \times T$ with a $T \times n$ matrix, to obtain a $n \times n$ matrix. The multiplication operation is quadratic is then $O(nT^2)$, and the inversion operation is $O(n^3)$. Hence, when either $T, n$ are large, matrix inversion can take a long time.

In particular, when $T$ is large, then gradient descent may give you a very good solution in a shorter time than it would take to do the matrix computations.

# Some matrix algebra reminders

## The identity matrix $\boldsymbol{I} \in \mathbb{R}^{n \times n}$

- For this matrix, $I_{i,i} = 1$ and $I_{i,j} = 0$ when $j \neq i$.
- $\boldsymbol{I}\boldsymbol{x} = \boldsymbol{x}$ and $\boldsymbol{I}\boldsymbol{A} = \boldsymbol{A}$.

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## The inverse of a matrix $\boldsymbol{A} \in \mathbb{R}^{n \times n}$

$\boldsymbol{A}^{-1}$ is called the inverse of $\boldsymbol{A}$ if

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- or equivalently $\boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}$.

## The pseudo-inverse of a matrix $\boldsymbol{A} \in \mathbb{R}^{n \times m}$

- $\tilde{\boldsymbol{A}}^{-1}$ is called the left pseudoinverse of $\boldsymbol{A}$ if $\tilde{\boldsymbol{A}}^{-1}\boldsymbol{A} = \boldsymbol{I}$.

$$\tilde{\boldsymbol{A}}^{-1} = (\boldsymbol{A}^\top \boldsymbol{A})^{-1}\boldsymbol{A}^\top, \qquad n > m$$

- $\tilde{\boldsymbol{A}}^{-1}$ is called the right pseudoinverse of $\boldsymbol{A}$ if $\boldsymbol{A}\tilde{\boldsymbol{A}}^{-1} = \boldsymbol{I}$.

$$\tilde{\boldsymbol{A}}^{-1} = \boldsymbol{A}^\top (\boldsymbol{A}\boldsymbol{A}^\top)^{-1}, \qquad m > n$$

The Linear Model
- Simple linear regression
- History
- Multiple linear regression

Solution methods
- Gradient Descent
- Least-Squares

Regression libraries in Python
- sklearn
- statsmodels

Problems
- Interpretation Problem parameters
- Exercises

# sklearn

## Fitting a model to data

```
from sklearn.linear_model import LinearRegression
model = LinearRegression().fit(X, Y)
```

## Getting predictions

We can get predictions for all inputs as an array

```
Z = model.predict(X)
```

# Statsmodels

## Fitting a model to data X, Y

```
import statsmodels.api as sm
Xa = sm.add_constant(X) # adds a constant factor to the data
model = sm.OLS(Y, Xa)
results = model.fit()
```

## Getting predictions

The prediction is not just a point!

```
z = results.get_prediction(Xa[t])
z.predicted_mean # This is E[y|x]
```

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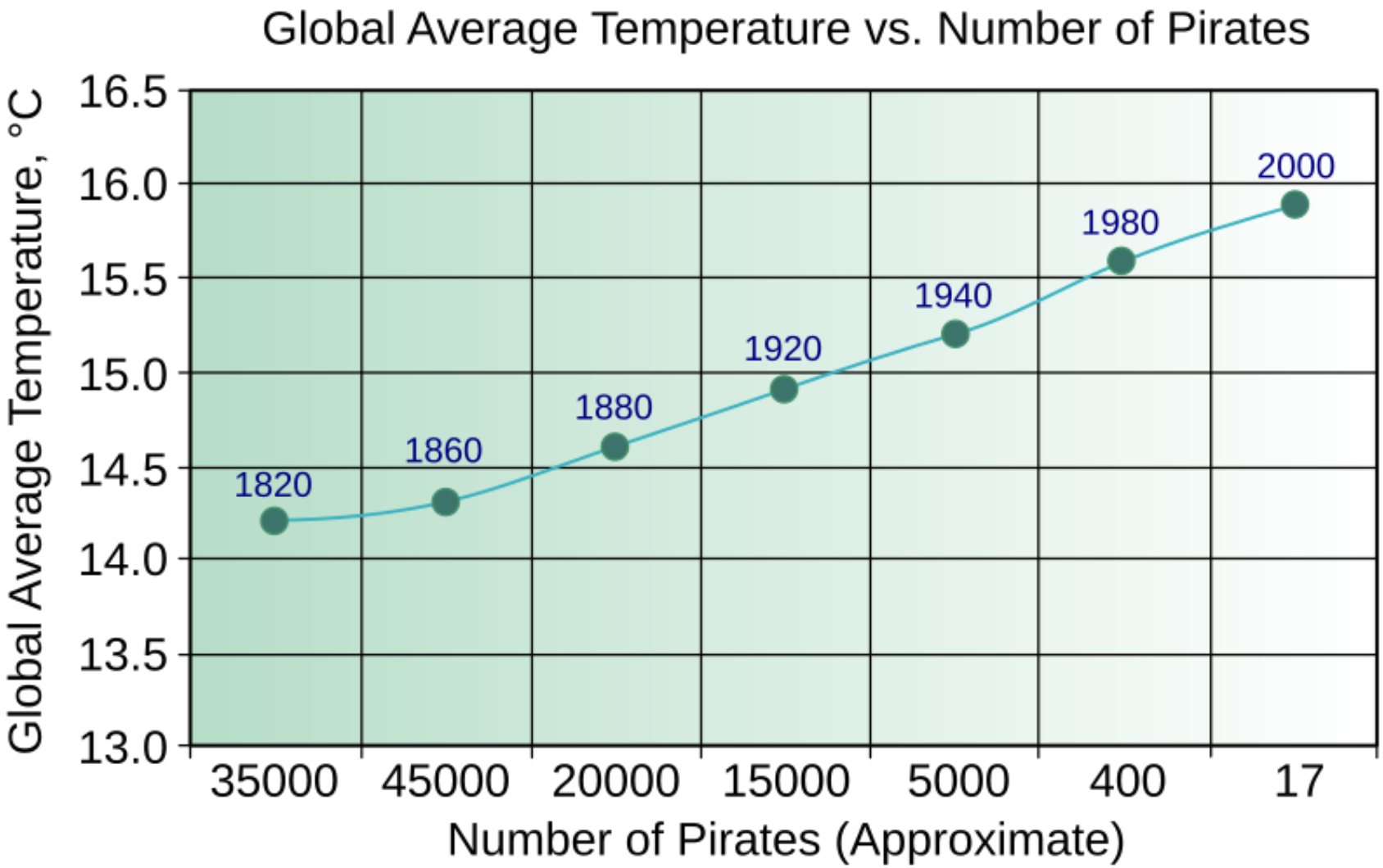
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# Pitfalls

- $\theta_i$ tells us how much $y$ is correlated with $x_{t,i}$
- However, multiple correlations might be evident.
- Some features may be irrelevant
- The relationship may not be linear
- Correlation is not causation

# Correlation is not causation



# Linear regression exercises

- Exercises 8, 13 from ISLP
- A variant of Ex. 13 but with Y generated independently of X.